

Profits, Prices, and the Power of Information

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Every trade in financial markets reflects a disagreement: the buyer believes the asset is undervalued, while the seller believes it's overvalued. Both sides cannot possibly be right, as that would lead to a contradiction, yet both sides willingly partake in this exchange. Clearly, there is some sort of information asymmetry here. The two individuals engaged in this trade either hold different information about the asset, or they interpret the same information in different ways, since if both parties held exactly the same facts and reached the same conclusion, a trade would never happen!

This simple chain of reasoning reveals a fundamental (yet often overlooked) truth about financial markets. On the surface, they may merely be a vehicle through which trades are conducted. But in reality, the financial market is an unbelievable machine that processes all the contradictory information backing our trades, and it outputs one number: a price. This market price mechanism plays a very important role in centralizing the dispersed information on which our market participants act.

An important aspect of the system is that the market doesn't magically know everything- it only integrates information after someone acts on it by trading. This causes a problem. If a person knows something nobody else on Earth does, why would she make her knowledge public? If she doesn't act on her knowledge, the market will never incorporate it into the price, and hence the price will never reach its value implied by available information. The market solves this problem in a very simple way.



It offers profit as the reward for contributing accurate information to the pricing system. Now you can either keep your information secret and do nothing or reveal it and make a profit if it turns out to be accurate. This essentially makes profit the opportunity cost of keeping your information private. With nothing to gain from secrecy and the huge opportunity cost of potential lost profit, every rational agent would rather reveal their private information by trading in the market, thus making the market more efficient by aggregating more information into its price.

Collecting the accurate information that this price mechanism relies on is rarely free; it takes time, effort, money and skill to gather this information. If nobody had an incentive to collect this information, they would never do it since it'd be a net loss for them. Without profits, the markets would base prices off far less information, and thus prices may be quite far away from their 'fair' value. Thus, profits are far from something to be vilified- they are a crucial aspect in making markets fairer for all by bringing prices closer to their information-reflecting value. In this manner, profits also serve as a feedback mechanism by rewarding those who contribute new and accurate information (by moving prices closer to their "fair" value) while penalizing those who contribute noise with losses.

Consider a situation where you own 50 shares of a stock priced at INR 1000 on exchange A, and find the same stock listed at INR 1005 on exchange B. Here you are provided with two choices; you can either utilize that information about the market to make a profit off of it, or you can decide not to come forward with that information. If you decide to go ahead with the first option, you will essentially be partaking in the practice of standard arbitrage, wherein the existence of different prices on different exchanges is utilized smartly to make a profit. By doing so, you will be sharing your information and thereby incorporating it into the stock prices. Buying the stock on exchange A increases demand on the exchange, and selling it on B increases supply in exchange B. This pulls the price on exchange A up, while lowering the price on exchange B. Your act of arbitrage then helps the prices across the two exchanges converge towards their information-implied value. If you decide against using your knowledge here, it creates a lose-lose situation- you lose out on profits, and the markets lose out on information, and thus the stocks trade at a deviation from their optimal prices.

You might think that since profit drives prices closer to their implied value, enough profit would eventually lead us exactly to the perfectly efficient market price.



But here's the twist: economists Sanford Grossman and Joseph Stiglitz showed that perfect efficiency actually destroys itself. If a perfect informationally efficient market existed, you would never make a profit since you could never contribute any new information. If you lose the profit incentive, you would never go through the pain of analyzing company cash flows or balance sheets to acquire information before making a trade. And if nobody takes part in information acquisition, prices become relatively inefficient since they're based on less information than before, which contradicts the initial assumption that the market is efficient and hence the paradox. The conclusion is then straightforward: some degree of informational inefficiency must exist for markets to function via the profit incentive.

Inefficiency being a necessary condition for markets to incentivize information gathering raises an interesting question: how far will market participants go to exploit these informational inefficiencies? Well, they'll go as far as they can because in financial markets, even a small edge could potentially be worth millions. One of the first people to recognize the importance of acquiring new information was Jim Simons, a former codebreaker at the NSA and a mathematics PhD from MIT. He founded Renaissance Technologies

(RenTech) and created the legendary Medallion Fund based on the idea of going as far as possible to find new information about stocks. His team used advanced mathematical models to identify trends and relations in prices that nobody else in the 1980/90s could identify. He was so adamant on hiring people with fresh ideas that he once famously said, "We never hired anyone from the financial world at Renaissance. We never did. Because they didn't have anything to add... We hire people who have done good science."

His unique approach and obsession with finding patterns in stock prices netted him handsome rewards. Since its inception, the Medallion fund has generated average annual returns of 66% before fees and 39% after, racking up trading gains of more than \$100 billion. No one in the investment world comes close. Warren Buffett, George Soros, Peter Lynch, Steve Cohen, and Ray Dalio— they all fall short. Today, Simon's approach is widely adopted by many new 'quant' trading firms and even legacy institutions like Barclays and Goldman Sachs. This pursuit of information has also clearly made some investors legends, with Jim Simons showing what happens when new insights genuinely improve price discovery. But when pushed too far, this pursuit of information may step into the realm of unethical practices. In order to gain an information edge, participants may try to obfuscate

information from price signals, and the Jane Street case is a clear example. SEBI argued that the firm's trades created a false or misleading appearance of market activity. Their claim was that the prices in bank stocks appeared to be rising on genuine demand, but were in fact an engineered move that misled investors. Retail traders, acting on these deceptive signals, bought at higher prices only to face losses when Jane Street dumped their shares and prices collapsed.

Over two years, this move allegedly earned Jane Street \$4.3 billion. But unlike the profit that comes from contributing genuine information to the market, these gains came from injecting noise into the pricing mechanism. By creating a false appearance of demand, they weakened the market's central function, which is using prices as an aggregator of information. In this case, the huge profits to be made no longer incentivized Jane Street to contribute information to prices, but instead to exploit investors' trust in the price mechanism. This inversion of the incentive system is what makes the Jane Street case so striking. The very mechanism that usually aligns private profit with public good was turned upside down. It shows us that although inefficiency is necessary for markets to function, it can also be exploited to manipulate the market.

Financial markets are essentially paradoxical: efficiency requires inefficiency, and markets require both freedom and discipline. Their strength lies in the way profit incentivizes traders to reveal and act on private information, thereby aggregating it into prices and making markets fairer for everyone. Whether that information drives innovation or manipulation depends on the guardrails we set, and on the values of those who trade within them. If Renaissance Technologies represents the upside of inefficiency, Jane Street illustrates its dark reflection. Both remind us that profit is not inherently virtuous or corrupt; it is simply the fuel that powers the system. Adding this fuel to the price mechanism helps financial markets live up to their reputation as machines that process information and guide prices - but only if society can find ways to align private profit incentives with the public good.